Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/07/2025
PI Name:	Chlebek, Carolyn Ph.D.		
Project Title:	Bone Metabolism is Altered by Skeletal Unloading and Nutrient Limitation During Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Food and Nutrition**:Risk of Performance Decrement and Crew Illness Due to Inadequate Food and Nutrition		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2024 TRISHRFA2401PD: Translational Research Institute for Space Health (TRISH) Announcement Soliciting Postdoctoral Fellowship Applications
Start Date:	09/01/2024	**End Date:**	07/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bouxsein, Mary Ph.D. (Beth Israel Deaconess Medical Center, Inc.) Rosen, Clifford Ph.D. (MENTOR: MaineHealth)		
Grant/Contract No.:	NNX16AO69A-P0901		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Long-duration spaceflight reduces bone quality via early onset osteoporosis. During spaceflight, the reduced gravitation loading on the skeleton contributes to this bone loss in astronauts. Additionally, long-duration spaceflight is associated with inadequate food and nutrition due to limited spacecraft storage and altered astronaut appetite, which can negatively alter bone health, independent of loading. Reducing nutrient intake via calorie restriction decreases bone quality and increases fracture risk. Both mechanical loading and caloric restriction individually affect bone metabolism. In various tissues, cellular metabolism is altered by spaceflight; however, the bioenergetic state of bone tissue in microgravity is unknown. Metabolism may be the conduit by which combined unloading and caloric restriction drives bone loss. The metabolic pathways altered within the skeleton during both mechanical unloading and nutrient limitation could serve as dietary or therapeutic targets to protect bone health during long-duration spaceflight. To determine the synergism between mechanical unloading and nutrient limitation, we will initiate both conditions in mice for 4 weeks. Mice will have reduced skeletal loading via hindlimb tail suspension, be nutrient-limited via caloric restriction, or receive a combination of both treatments. Control animals will be included. Following 4 weeks of treatment, we will determine changes in bone morphology and the metabolic state of bone via carbon tracing. We also will examine cellular metabolism within the most abundant bone cell, the osteocyte. Using both in vivo and in vitro investigations, we will determine osteocytic metabolic needs during bone maintenance and mechanotransduction. This research will define the metabolic state of bone during periods of unloading and nutrient limitation, and will elucidate the critical role of specific metabolic pathways in bone maintenance and mechanotransduction.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal will expand our knowledge of bioenergetic function and interrogate additional mechanisms of spaceflight-induced bone loss. The experiments proposed are the first to address the synergism between nutrient limitation and unloading, better representing the conditions faced by astronauts during long-duration spaceflight missions. The proposed experiments use novel metabolic tracing methods that will allow us to identify aberrant metabolic pathways activated during caloric restriction and hindlimb unloading. Using both genetic models and in vitro experiments, our novel interrogation of osteocyte-specific metabolism during bone maintenance and mechanotransduction will enhance our understanding of bone cell bioenergetics. Importantly, this proposal will yield compartment-specific investigations, potentially spurring the design of trabecular- and cortical-specific interventions. The influence of metabolism on bone health is underexplored, despite evidence that successful osteogenic therapeutics alter bioenergetic function. This work will set the stage for future studies that directly target bone metabolism through dietary or therapeutic interventions to enable long-duration spaceflight without bone loss.
Task Progress:	Long-duration spaceflight reduces bone quality via early onset osteoporosis. Multiple factors may contribute to bone loss during long-duration spaceflight. During spaceflight, the reduced gravitation loading on the skeleton causes bone loss in astronauts; however, the added factors of caloric restriction and social isolation may exacerbate bone loss during long-duration spaceflight. Reducing nutrient intake via calorie restriction decreases bone mass and increases fracture risk in both rodent models and in humans. The skeletal effects of social isolation in humans are unknown, but in mice, chronic stress through social isolation causes bone loss in males, but not females. Understanding the impact of caloric restriction and social isolation on bone loss in astronauts is critical, as long-duration spaceflight is associated with inadequate food and nutrition due to limited spacecraft storage and altered astronaut appetite. In addition to experiencing calorie deficits astronauts also undergo social isolation due to small crew sizes. This proposal seeks to understand how combined mechanical unloading, caloric restriction, and social isolation may contribute synergistically to bone loss and the biological mechanisms at play. To address our aims, we use mouse models to understand the changes to bone morphology and biological activity. We first sought to determine if bone loss due to social isolation and caloric restriction is additive. We hypothesized that calorie restriction would reduce bone mass in both male and female mice. Based on previous studies, we expected that in males but not females, social isolation would further worsen the deficits to bone incurred by caloric restriction. For these experiments, we used the genetically-identical inbred C57BL/6J mice, which have established responses to both caloric restriction and social isolation. We randomly assigned mice to one of four interventions: (A) group housing with ad libitum diet, (B) single housing with ad libitum diet, (C) group housing with caloric restriction, or (D) single housing with caloric restriction. Interventions were employed from ages 16 weeks to 24 weeks in both sexes. At euthanasia, calorie-restricted mice had less lean and fat mass compared to mice consuming an ad libitum diet. Within calorie-restricted animals, socially isolated mice lost more lean mass compared to animals that were group housed. As anticipated, the combination of social isolation and caloric restriction produced additive negative effects to trabecular bone in males; following calorie restriction, trabecular thickness was reduced in socially isolated, but not group-housed, mice. As hypothesized, social isolation reduced both cortical and trabecular bone mass in male mice. Calorie restriction, regardless of housing, reduced trabecular number in males. Cortical bone area was not reduced with diet or isolation, but social isolation increased the marrow area in male mice, resulting in reduced cortical thickness. As expected, housing did not affect bone mass in female mice. Regardless of housing, in females, calorie restriction increased trabecular bone volume fraction and trabecular but reduced trabecular thickness. Cortical area, cortical thickness, and maximum moment of inertia also were reduced by calorie restriction in females. Analysis of circulating factors revealed that changes to both bone formation and bone resorption may contribute to bone loss during isolation and calorie restriction. P1NP, a protein released into the bloodstream during bone formation, was reduced in socially-isolated, calorie-restricted animals. In both sexes, TRAcP 5b, a protein associated with bone resorption, was increased with social isolation but not altered by calorie restriction. Building on this work, in Year 2, we will use the hindlimb unloading model to assess the combined effects of mechanical unloading, calorie restriction, and social isolation on the skeleton. During the first year, we also established mouse colonies that will allow us to understand the individual roles of pyruvate-mediated oxidative phosphorylation and glycolysis in bone maintenance. We are targeting these pathways in the most abundant bone cell, the osteocyte, by using the DMP1-Cre promoter. To examine pyruvate-mediated oxidative phosphorylation in osteocytes, we are conditionally deleting Mpc2 expression in osteocytes by mating the floxed Mpc2 mouse (Mpc2fl/fl) with our DMP1-Cre mice, resulting in Mpc2OcyKO animals. In Mpc2OcyKO mice, pyruvate transportation into the mitochondria will be inhibited only in osteocytes. Glycolysis will be evaluated by conditional deletion of Ldha expression using the floxed Ldha mouse (Ldhafl/fl) mated to our DMP1-Cre mice, resulting in LdhaOcyKO animals. The last step of anaerobic glycolysis will be inhibited only in osteocytes in LdhaOcyKO mice. Breeding is currently underway and will produce our desired Mpc2OcyKO or LdhaOcyKO mice, as well as littermate controls for both knockout models. Once a sufficient number of mice are produced within the Mpc2OcyKO or

	LdhaOcyKO colonies, we will evaluate bone morphology and strength. Additionally, we plan to evaluate osteocyte mechanotransduction using in vitro fluid shear experiments. Using the immortalized osteocyte cell line OCY454 cells, we will therapeutically inhibit pyruvate-mediated oxidative phosphorylation and glycolysis, then measure mechanoresponsiveness. To inhibit pyruvate-mediated oxidative phosphorylation, UK5099 will be delivered to the cells. To assess glycolytic contributions to mechanoresponsiveness, 2-deoxyglucose (2DG) will be administered to prevent the conversion of glucose to glucose-6-phosphate. We will apply fluid shear, then evaluate the number of responding osteocytes. These experiments will help us understand the metabolic mechanisms employed by osteocytes during both bone growth and mechanotransduction.
Bibliography Type:	Description: (Last Updated: 11/07/2025)
Articles in Other Journals or Periodicals	Chlebek C, Bussberg V, Narain NR, Kiebish MA, Tseng YH, Lynes MD. "Baseline serum metabolites predict fractures in a subset of accord participants who were black." SSRN preprint server. Posted June 26, 2025. http://dx.doi.org/10.2139/ssrn.5297235 , Jun-2025
Awards	Chlebek C. "American Society for Bone and Mineral Research Harold M. Frost Young Investigator Award. " Jun-2025